

Section 1 → BRD

- 1. Business Problem**
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- **Business Problem:**

The pharmacy is experiencing inventory management challenges that lead to financial losses and operational inefficiencies.

Current issues include:

1. Overstocking of slow-moving products, resulting in increased holding costs and higher risk of product expiration.
2. Stock shortages of fast-moving medications, leading to missed sales opportunities and reduced customer satisfaction.
3. Financial losses due to expired medications.
4. Lack of accurate forecasting when placing purchase orders, causing pharmacists to rely on manual judgment.

The pharmacy requires a system that analyzes historical sales data and inventory movements to provide intelligent stock recommendations, reduce expiries, and improve product availability.

- **Business Objectives:**

BO-001

Reduce inventory losses caused by expired medications by providing early expiry alerts, near-expiry reporting, and expiry monitoring capabilities.

BO-002

Improve product availability by ensuring critical medications remain in stock.

BO-003

Reduce stockout incidents by at least 40% through automated reorder recommendations.

BO-004

Optimize inventory investment by minimizing overstocking of slow-moving products.

BO-005

Improve inventory planning accuracy by replacing manual reorder decisions with data-driven recommendations.

BO-006

Increase operational efficiency by providing pharmacists with automated daily replenishment reports.

BO-007

Provide management with real-time visibility into inventory performance through dashboards and analytical reports.

BO-008

Support informed inventory decision-making by analyzing product movement trends and sales performance.

BO-009

Improve inventory control by maintaining accurate batch-level tracking for quantities and expiry dates.

BO-010

Increase accountability and compliance by recording inventory-related actions and approval activities in the audit log.

BO-011

Enable proactive management of inventory risks by identifying near-expiry and expired products before they impact business performance.

BO-012

Improve branch-level inventory governance by implementing an approval workflow for modified reorder quantities.

- **Success Metrics**

SM-001

Reduce expired inventory value by at least 30% within six months of system implementation.

SM-002

Reduce stockout occurrences by at least 40% within six months of system implementation.

SM-003

Improve reorder decision accuracy by utilizing automated recommendation calculations.

SM-004

Reduce manual inventory review effort through automated replenishment and expiry reports.

SM-005

Provide 100% visibility of products approaching expiry through dashboard monitoring and reporting.

- **Stakeholders:**

Primary Stakeholders

- 1- Pharmacist
- 2- Branch Manager

Secondary Stakeholders

- 1- Pharmacy Owner
 - 2- External POS System
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- **Scope:**

*** In Scope**

1. Product Management
2. Batch Management
3. Purchase Invoice Management
4. Inventory Tracking
5. Expiry Tracking
6. Expiry Alerts
7. Near Expiry Reporting
8. Reorder Recommendations
9. Daily Replenishment Report
10. Inventory Dashboard
11. Sales Dashboard
12. Fast-Moving Product Analysis
13. Slow-Moving Product Analysis
14. Reorder Approval Workflow
15. Audit Logging

Out of Scope

1. HR Management
 2. Payroll Management
 3. Accounting Management
 4. Supplier Management
 5. Warehouse Management System
 6. Procurement System
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- **Business Rules**

BR-001

Every product must have a unique Product ID.

BR-002

A product cannot be created without mandatory product information.

BR-003

Each batch must belong to an existing product.

BR-004

Each product may have multiple batches, but duplicate batch numbers for the same product shall update the existing batch quantity instead of creating a new batch.

BR-005

Expired batches cannot be added to the system.

BR-006

Expiry alerts shall be generated when a batch has 90 days or less before expiry.

BR-007

The system shall generate reorder recommendations during the daily replenishment process when stock $\leq$ threshold.

BR-008

The system shall use the previous 6 months sales history when calculating reorder recommendations.

BR-009

The system shall consider the current stock quantity when calculating reorder recommendations.

BR-010

The system shall consider product expiry risk when calculating reorder recommendations.

BR-011

The system shall consider product movement classification when calculating reorder recommendations.

BR-012

The system shall generate reorder recommendations that cover 30 days of expected demand.

BR-013

If the calculated quantity is less than or equal to zero, the system shall not generate a reorder recommendation.

BR-014

The system shall add every product reorder recommendation to The Daily Replenishment Report.

BR-015

The system shall generate the Daily Replenishment Report start of each business day.

BR-016

Pharmacists may modify recommended reorder quantities before approval.

BR-017

A justification is mandatory whenever a pharmacist modifies a recommended reorder quantity.

BR-018

Modified Daily Replenishment Report quantities require Branch Manager approval.

BR-019

The Branch Manager may:

- * Approve pharmacist quantity
- * Approve original recommendation
- * Enter custom quantity
- * Reject

BR-020

All reorder actions shall be recorded in the audit log.

BR-021

Products shall remain in the Daily Replenishment Report until action is completed.

BR-022

The system shall classify products based on average 3 months sales:

Fast > Configurable value

Medium >= Configurable value and < Configurable value

Slow < Configurable value

BR-023

Products threshold shall be configurable by the Pharmacist and it must be greater than zero.

BR-024

Recommended Quantity shall be measured by Base Quantity and adjusted by Expiry Risk.

Base Quantity =

(Expected 30-Day Demand – Current Stock)

- IN High Risk Of Expiry

Recommended Quantity = Base Quantity x 70%

- IN Medium Risk Of Expiry

Recommended Quantity = Base Quantity x 90%

- IN Low Risk Of Expiry

Recommended Quantity = Base Quantity

BR-25

A batch shall automatically be marked as Near Expiry when it has 90 days or less before expiry.

BR-026

A batch shall automatically be marked as Expired at the start of the day following its expiry date.

BR-027

Expired batches shall not be included in reorder calculations.

BR-028

Disposed quantities shall be deducted from inventory automatically.

BR-029

Disposed batches shall remain in the system for audit and reporting purposes.